THE UNIVERSITY OF MELBOURNE

COMP30023 – Computer Systems

Internet (Network) Layer – NAT and Subnets and Fragmentation

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Recap

Internet Control Message Protocol

- Ping
- Traceroute

Dynamic Host Configuration Protocol (DHCP)

- Assigns IP addresses and other things

Address Resolution Protocol (ARP)

- Finding layer 2 (MAC) address to match IP

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Summary

- Network Address Translation (NAT)
- Subnets
- Fragmentation
  - IPv4
  - IPv6 if there is time

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IPv4 address scarcity

- As IP addresses became scarce, methods for handling many more clients were developed
- While IPv6 would solve the problem, a stop gap was needed
- Private addresses
  - Many hosts in a company only need internal access
  - “Private” subnets 192.168.0.0/16, 172.16.0.0/12, 10.0.0/8
  - Can be reused: unique address within organisation, not globally
- Intention: “Application layer proxies” to access outside services
  - Instead…

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Network Address Translation (NAT)

- Each customer/home is assigned one public IP address
  - Businesses might be issued a few
- Internally hosts/interfaces are issued Private IP addresses
  - Recall 10.0.0.0/8 - 10.255.255.255 (as an example)
- Internal IP addresses are used for communicating among hosts in the Local Area Network (LAN)
- They must never be used on the public internet
- When a packet is heading out of the network (to the ISP) the internal address is translated to the public IP address

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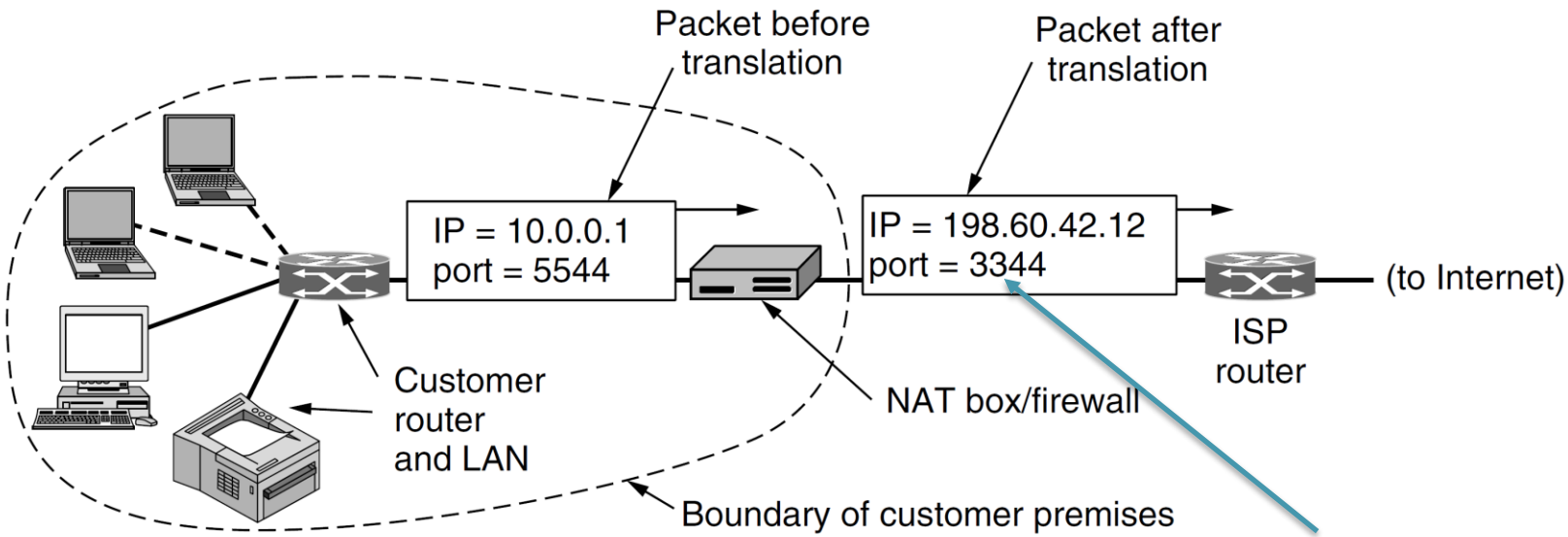
Network Address Translation (NAT)



Port number also changes → NAPT

TN 6th 5-56

Network Address Translation (NAT)

- How NAT works:
  - Assumes TCP/UDP (some exceptions), in particular the locations of source and destination port fields
  - NAT box replaces IP source address (10.x.y.z) with public IP address
  - TCP source port replaced with index of entry in NAT translation table
    - One of 65,536 entries (16 bits – same as TCP port field)
    - Each entry contains original IP address (private IP) and original source port number
  - IP and TCP checksums are recalculated
  - When a packet arrives from the internet at the NAT box it looks up the destination port from the TCP header in the translation table
    - Retrieves original source port and source IP address, updates headers and checksums and sends to the internal host

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Network Address Translation (NAT)

- Criticisms of NAT
  - Breaks end-to-end connectivity: an interface in the private network can only receive packets once it has sent packets out and created a mapping (some exceptions)
  - Violates IP architectural model that states every IP address indicates a unique interface on the internet (thousands of interfaces connecting to the internet have 10.0.0.1)
  - "Layering violation" by assuming nature of payload contents – initially only worked for TCP and UDP. Must snoop on FTP messages: https://enterprisedt.com/products/edtftpjssl/doc/manual/html/howtoftpthroughafilewall.html
  - Changes internet from connectionless to pseudo-connection-oriented
    - NAT maintains connection state, if it crashes all connections are lost
  - Limits number of connection, since port numbers are 16 bits.

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Network Address Translation (NAT)

- Despite criticisms, it is widely deployed, particularly in homes and small businesses
  - Carrier grade NAT: ISP only gives customers private addresses
- Significant security advantage
  - Since packets can only be received once an outgoing connection has been created, the internal network is greatly shielded from attacks from incoming unsolicited packets
  - NAT should not replace firewalls
- Likely to remain in use even after IPv6 is widely deployed and there is no longer a scarcity of IP addresses

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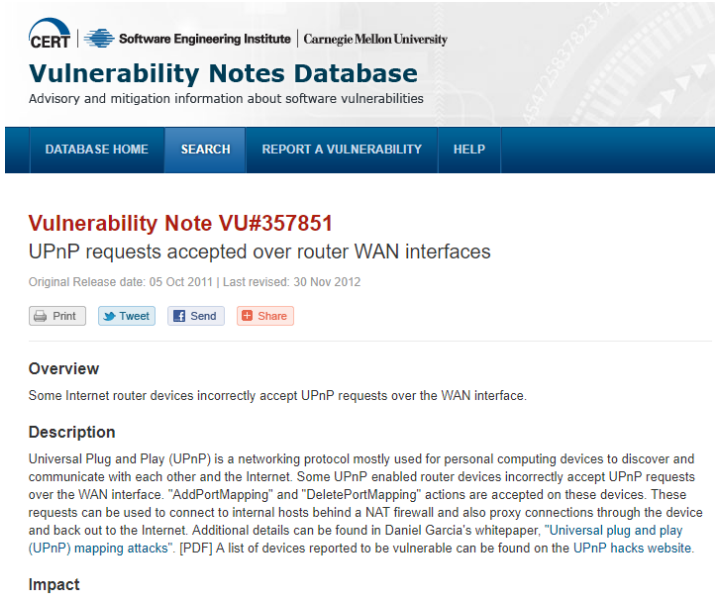
And finally…

- Attempts have been made to overcome the weaknesses of NAT
- Notably, Universal Plug and Play (UPnP)
- As internet use grew, the need for home machines to act as servers grew
  - Gaming, chat, media players, etc.
- UPnP implements the Internet Gateway Device Protocol that allows port mappings to be created in the NAT translation table to allow servers to be run on the internal network
- Bad implementations of UPnP have allowed attackers to alter translation tables from external IP addresses



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Vulnerability Note VU#357851

UPnP requests accepted over router WAN interfaces

Original Release date: 05 Oct 2011 | Last revised: 30 Nov 2012

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Overview

Some Internet router devices incorrectly accept UPnP requests over the WAN interface.

Description

Universal Plug and Play (UPnP) is a networking protocol mostly used for personal computing devices to discover and communicate with each other and the Internet. Some UPnP enabled router devices incorrectly accept UPnP requests over the WAN interface. "AddPortMapping" and "DeletePortMapping" actions are accepted on these devices. These requests can be used to connect to internal hosts behind a NAT firewall and also proxy connections through the device and back out to the Internet. Additional details can be found in Daniel Garcia's whitepaper, "Universal plug and play (UPnP) mapping attacks". [PDF] A list of devices reported to be vulnerable can be found on the UPnP hacks website.

Impact

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Acknowledgement

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- Some of the images included in the notes were supplied as part of the teaching resources accompanying the text books listed in lecture 1.
  - (And also) Computer Networks, 6th Edition, Tanenbaum A., Wetherall. D. https://ebookcentral.proquest.com/lib/unimelb/detail.action?docID=6481879
- Textbook Reference:

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